

Surgery of the Ankle

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[Instructions for Use](#)

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Related Commercial/Individual Exchange Policy

- [Omnibus Codes](#)
- [Outpatient Surgical Procedures – Site of Service](#)

Application

UnitedHealthcare Commercial

This Medical Policy applies to UnitedHealthcare Commercial benefit plans.

UnitedHealthcare Individual Exchange

This Medical Policy applies to Individual Exchange benefit plans.

Coverage Rationale

Surgery of the ankle is proven and medically necessary in certain circumstances. For medical necessity clinical coverage criteria, refer to the InterQual® CP: Procedures:

- Arthrodesis, Ankle (Talotibial Joint)
- Arthroscopy, Surgical, Ankle
- Arthrotomy, Ankle
- Total Joint Replacement (TJR), Ankle

[Click here to view the InterQual® criteria.](#)

Osteochondral allograft or autograft transplant is unproven and not medically necessary for treating cartilage defects of the ankle due to insufficient evidence of efficacy.

Medical Records Documentation Used for Reviews

Benefit coverage for health services is determined by the member specific benefit plan document and applicable laws that may require coverage for a specific service. Medical records documentation may be required to assess whether the member meets the clinical criteria for coverage but does not guarantee coverage of the service requested; refer to the guidelines titled [Medical Records Documentation Used for Reviews](#).

Applicable Codes

The following list(s) of procedure and/or diagnosis codes is provided for reference purposes only and may not be all inclusive. Listing of a code in this policy does not imply that the service described by the code is a covered or non-covered health service. Benefit coverage for health services is determined by the member specific benefit plan document and applicable laws that may require coverage for a specific service. The inclusion of a code does not imply any right to reimbursement or guarantee claim payment. Other policies and guidelines may apply.

CPT Code	Description
Arthrodesis, Ankle (Talotibial Joint)	
29899	Arthroscopy, ankle (tibiotalar and fibulotalar joints), surgical; with ankle arthrodesis
Arthroscopy, Surgical, Ankle	
29891	Arthroscopy, ankle, surgical, excision of osteochondral defect of talus and/or tibia, including drilling of the defect
29892	Arthroscopically aided repair of large osteochondritis dissecans lesion, talar dome fracture, or tibial plafond fracture, with or without internal fixation (includes arthroscopy)
29894	Arthroscopy, ankle (tibiotalar and fibulotalar joints), surgical; with removal of loose body or foreign body
29895	Arthroscopy, ankle (tibiotalar and fibulotalar joints), surgical; synovectomy, partial
29897	Arthroscopy, ankle (tibiotalar and fibulotalar joints), surgical; debridement, limited
29898	Arthroscopy, ankle (tibiotalar and fibulotalar joints), surgical; debridement, extensive
29899	Arthroscopy, ankle (tibiotalar and fibulotalar joints), surgical; with ankle arthrodesis
Arthrotomy, Ankle	
27685	Lengthening or shortening of tendon, leg or ankle; single tendon (separate procedure)
28446	Open osteochondral autograft, talus (includes obtaining graft[s])
Total Joint Replacement (TJR), Ankle	
27702	Arthroplasty, ankle; with implant (total ankle)

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Description of Services

Osteoarthritis is also known as degenerative arthritis and is common for many people after they reach middle age; however, it may occur in younger people as well. In osteoarthritis, the cartilage in the joint gradually wears away. As the cartilage wears away, it becomes frayed and rough, and the protective space between the bones decreases. This can result in bone rubbing on bone and produce painful osteophytes (bone spurs).

Posttraumatic arthritis can develop after an injury to the foot or ankle, and dislocations and fractures are the most common injuries that lead to posttraumatic arthritis. Like osteoarthritis, posttraumatic arthritis causes the cartilage between the joints to wear away and can develop many years after the initial injury.

Clinical Evidence

Osteochondral Allograft or Autograft Transplant

There is insufficient quality evidence regarding the safety and efficacy of osteochondral allograft or autograft transplant. Future studies, including randomized controlled trials with comparison groups, are needed, along with long-term results.

Wen et al. (2025) conducted a PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)–registered systematic review under PROSPERO, evaluating the long-term clinical efficacy and safety of surgical treatments of osteochondral lesions of the talus (OLTs), with a minimum 5-year follow-up. Studies included adults aged over 18 years from prospective or retrospective cohort, case-control, or case series studies with more than five individuals. There were 43 studies, with 2,721 individuals (62.2% male) aged 23.9 to 47.7 years and who had defect sizes of 0.9 to 29.1 cm² included in the review, which evaluated a total of 13 joint-preservation techniques and included follow-up durations of 5 to 24.1 years. The outcomes assessed included patient-reported outcomes, return to activity, complications, and failures. All studies that were included were non–randomized controlled trials and level IV evidence.

The primary outcome, patient-reported improvement in American Orthopaedic Foot and Ankle Society (AOFAS) scores, was significant ($p < 0.05$) across most studies, with preoperative scores of 40.4 to 89.4 and postoperative scores of 75.3 to 95.3, depending on technique; survival, defined as absence of revision surgery, ranged from 53.8% to 100% survival and 92.72% in a pooled analysis. Osteochondral allograft or autograft transplant had the highest range of complications (3.7%-59%). The authors concluded that the most common technique used was bone marrow stimulator (BMS), primarily for smaller lesions, and that it produced comparable survival rates compared with all other techniques. Additionally, they concluded that treatments should be personalized and evidence based and should consider lesion characteristics and patient factors. Limitations include the high heterogeneity across studies due to wide lesion size ranges and characteristics of the lesion. Additionally, the majority of studies were noncomparative in design and level IV evidence, which limits the strength of the findings and prevents meta-analysis of pooled data; several studies that were included had small sample sizes; objective measures used for patient-reported outcomes were lacking; concomitant or previous procedures were not controlled for; and baseline demographics were not often stratified into subgroups, which limited comparability and the conduct of a meta-analysis.

Correia Cardoso et al. (2024) conducted a systematic review that evaluated the operative treatment of nonprimary OLTs. The review included 50 studies, which involved 806 ankles of 794 individuals. The majority of the studies were retrospective in design, and only 14% included a comparison group. The Oxford Centre for Evidence-Based Medicine was used and showed that 86% of the studies were classified as level 4 evidence. All studies exhibited a high risk of bias in at least one domain, with 64% demonstrating a high risk across at least four domains. Cartilage substitution was the most common treatment, including 30% osteochondral autograft and 11% osteochondral allograft. Cartilage regeneration, comprising 37%, included 9% autologous chondrocyte implantation (ACI), 8% matrix-assisted autologous chondrocyte transplant, 2% bone marrow-derived cell transplant, and 17% autologous matrix-induced chondrogenesis (AMIC). Rescue procedures/HemiCAP and cartilage repair/BMS accounted for 14% and 9%, respectively. The AOFAS score was reported in 54% of the studies, and the mean improvement ranged from 13 to 57.8, with osteochondral autograft showing the most and least pronounced mean change. The pre- to postoperative meta-analysis revealed significant improvements across all treatments ($p < 0.05$), except for osteochondral allograft, which showed no significant improvement. The highest success rates were seen with ACI and osteochondral autograft, whereas HemiCAP had the lowest. The visual analog scale/numeric rating scale for pain score was reported in 48% of studies and showed mean improvements that ranged from 1.5 to 8, with osteochondral autograft displaying the most pronounced mean change and HemiCAP showing the lowest. The pre- to postoperative meta-analysis revealed significant improvements across most treatments ($p < 0.05$), with ACI demonstrating a large effect size, while osteochondral allograft and HemiCAP showed no significant improvements. The highest success rates were seen with ACI and osteochondral autograft, while HemiCAP had the lowest. The postoperative subjective satisfaction was reported in 46% of the studies, with a high rate of success across all operative treatments; AMIC and HemiCAP had the lowest rates. HemiCAP had the highest incidence of postoperative complications, and clinical failures were more frequent with AMIC (27%), osteochondral autograft (22%), osteochondral allograft (19%), and HemiCAP (28%). While ACI and osteochondral autograft showed improvements and higher success rates than HemiCAP and osteochondral allograft, the authors noted that despite the promising potential, the high risk of bias across the included studies warrants a cautious interpretation of these results. Limitations of the study include the high risk of selection and detection bias; retrospective nature; predominance of case series; variability in study designs and outcome measures; lack of comparison groups; use of nonvalidated outcome measures; and insufficient reporting of demographic variations and lesion characteristics.

Migliorini et al. (2022) conducted a systematic review to evaluate the efficacy of surgical management techniques for osteochondral defects. Surgical management techniques included osteochondral autograft, mosaicplasty, matrix-assisted autologous chondrocyte transplant, and AMIC. There were 13 articles included in the review, with a total of 521 procedures and a median length follow-up of 47.8 months (31.7-66.8 months). The authors noted that there was no difference between the treatment groups at baseline in terms of mean age, body mass index, individuals' sex, defect size, and visual analog scale and AOFAS scores. AMIC demonstrated the lowest rates of failure [log odds ratio (LOR), 0.94] and revision (LOR, 0.94), while osteochondral autograft evidenced the highest rates of failure (LOR, 3.48) and revision (LOR, 4.60). Limitations include the overall poor quality of many studies, with high selection bias due to a large number (10 of the 13 included studies) of retrospective comparative studies. Additionally, there were variances in the surgical approach, nature of the membrane, fixation methods, and location of the lesion.

Lambers et al. (2017) conducted a systematic review to identify the most effective surgical treatment for talar osteochondral defect after failed primary surgery. There were 21 studies included in the review, with a total of 299 individuals and 301 talar osteochondral defects. Of those studies, eight were retrospective case series, 12 were prospective case series, and one was a randomized controlled trial. Treatment strategies were divided into four groups: BMS (debridement and/or drilling), osteochondral transplant (autograft transfer, allograft transfer, and mosaicplasty), cartilage implantation (matrix-associated chondrocyte implantation and ACI), and chondrogenesis-inducing techniques (AMIC). BMS success percentages were 75% (debridement alone) and 69% (debridement and microfracture), with CIs of

47% to 91% and 42% to 87%. Osteochondral transplant was the most common procedure; however, a calculated success rate for all osteochondral transplant techniques combined was not possible because study designs varied for all studies. The authors were able to use a simplified pooling method, which resulted in a mean success rate of 90% (CI, 82%-95%) for the osteochondral autograft transfer procedure, 65% (CI, 46.2%-80.6%) for mosaicplasty, and 55% (CI, 39.7%-69.9%) for the osteochondral allograft transfer procedure. There was no significant difference between matrix-associated chondrocyte implantation and ACI, and the calculated success rate was 72% (CI, 56%-85%) and 59% (CI, 39%-77%). The success rates for AMIC were 67% (CI, 30%-90.3%) and 57% (CI, 32.6%-78.6%). Limitations of this study include the low methodological quality of the studies included; additionally, nearly half of the extracted data that were acquired through the direct approach of the authors limited the ability to collect all the variables desired, including complications, lesion size, and classification systems used. The authors noted that due to the low level of evidence and the limited number of individuals, a methodologically proper meta-analysis could not be completed, and it would be inappropriate to draw firm conclusions from the collected results. Further prospective investigations in a randomized comparative clinical setting are needed.

A Hayes (2013; updated 2015) Health Technology Assessment evaluated the osteochondral allograft transplant for articular disorders of the ankle. There were seven small, uncontrolled clinical studies that evaluated the safety and efficacy of osteochondral allografting for the treatment of OLTs or severe tibiotalar arthritis. Most of the studies were small, only one had a prospective design, and there were no randomized controlled trials identified. Four studies included individuals with OLTs, and two studies included individuals with ankle arthritis. The mean age range was 30 to 44 years, and follow-up times ranged from a mean of approximately 2 years to 5 years. Most studies showed a significant improvement in the mean scores on standardized instruments for pain and function following allografting, with an overall improvement rate in 52% to 89% of individuals; however, a considerable number of the allografts failed, requiring repeat allografting, arthrodesis, or arthroplasty (range, 11%-48%). The authors noted that the available evidence is insufficient to draw definitive conclusions regarding the safety, efficacy, and durability of osteochondral allograft transplant for articular ankle disorders. The overall quality of the evidence is low, with the existing studies having small numbers of individuals and being uncontrolled; additionally, the majority of studies were retrospective, making them prone to bias. Additional studies are needed that are controlled and that compare the long-term effectiveness and safety with those of other methods of restoring the articular cartilage. Other limitations include the differences in allograft preparation, surgical protocols; inconsistent reporting of radiological outcomes; and variability in outcome measures and follow-up times.

A Hayes (2012; updated 2014) Health Technology Assessment evaluated the osteochondral autograft or mosaicplasty for lesions of the talus. There were 10 available studies, which included one randomized comparative trial; one nonrandomized, prospective comparative study; six prospective studies; and two retrospective studies. The authors determined that there was insufficient evidence to support conclusions regarding the efficacy of the osteochondral autograft or mosaicplasty procedure in individuals with OLTs as well as the relative effectiveness compared with that of standard procedures. There was an overall low quality of evidence due to poor study design, small populations of individuals, variability in the measures used to assess outcomes, varying follow-up times, and substantial differences in surgical protocols and techniques. Additional well-designed trials that have long-term follow-up are needed to evaluate the effectiveness of surgical options for OLTs.

Clinical Practice Guidelines

American Orthopaedic Foot and Ankle Society (AOFAS)

The AOFAS (2022) published a position statement that supports the use of osteochondral autograft and allograft transplant for the treatment of OLTs that have failed nonsurgical management, especially for large-diameter lesions, cystic lesions, and lesions that have failed previous surgical treatment. The society considers osteochondral transplant to be a treatment option, with demonstrated improved outcomes maintained over long-term follow-up.

U.S. Food and Drug Administration (FDA)

This section is to be used for informational purposes only. FDA approval alone is not a basis for coverage.

Surgeries of the ankle are procedures and therefore are not regulated by the FDA. However, devices and instruments used during the surgery may require FDA approval. Refer to the following website for additional information: <http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfPMN/pmn.cfm>. (Accessed March 5, 2026)

References

American Orthopaedic Foot and Ankle Society (AOFAS). Position statement. The use of osteochondral transplantation for the treatment of osteochondral lesions of the talus. July 2022. Available at: <https://www.aofas.org/docs/default->

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Policy History/Revision Information

Date	Summary of Changes
07/01/2026	Related Policies - Added reference link to the Medical Policy titled <i>Outpatient Surgical Procedures – Site of Service</i> Supporting Information - Archived previous policy version 2026T0622N

Instructions for Use

This Medical Policy provides assistance in interpreting UnitedHealthcare standard benefit plans. When deciding coverage, the member specific benefit plan document must be referenced as the terms of the member specific benefit plan may differ from the standard plan. In the event of a conflict, the member specific benefit plan document governs. Before using this policy, check the member specific benefit plan document and any applicable federal or state mandates. UnitedHealthcare reserves the right to modify its policies and guidelines as necessary. This Medical Policy is provided for informational purposes. It does not constitute medical advice.

This Medical Policy may also be applied to Medicare Advantage plans in certain instances. In the absence of a Medicare National Coverage Determination (NCD), Local Coverage Determination (LCD), or other Medicare coverage guidance, CMS allows a Medicare Advantage Organization (MAO) to create its own coverage determinations, using objective evidence-based rationale relying on authoritative evidence ([Medicare IOM Pub. No. 100-16, Ch. 4, §90.5](#)).

UnitedHealthcare may also use tools developed by third parties, such as the InterQual® criteria, to assist us in administering health benefits. UnitedHealthcare Medical Policies are intended to be used in connection with the independent professional medical judgment of a qualified health care provider and do not constitute the practice of medicine or medical advice.